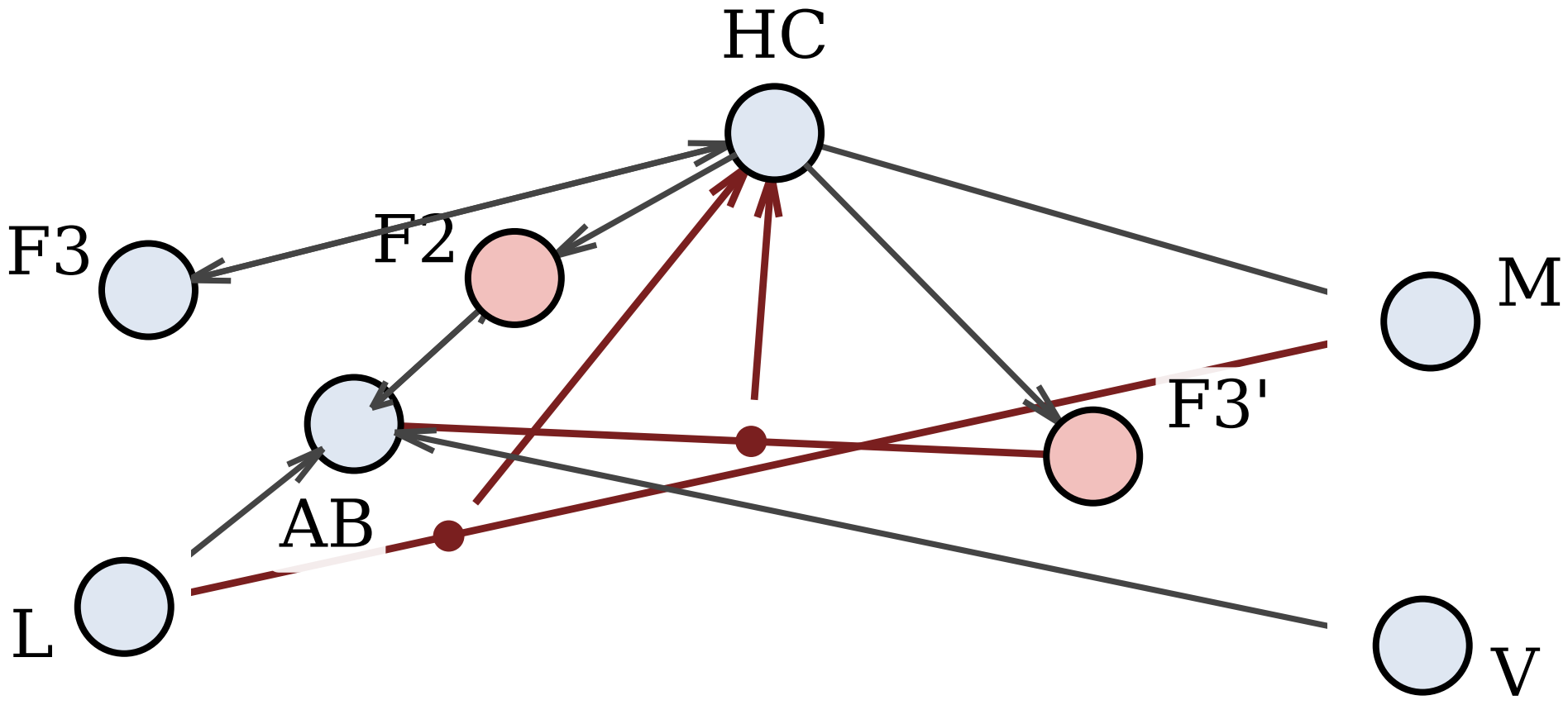The closure graph: HC $\Leftrightarrow$ (F2) $\wedge$ (F3′)